

Vedant Desai

(248) 704 – 4852 | vedantde@umich.edu | github.com/Verdent06 | linkedin.com/in/vedantde06

EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0

- **Coursework:** Data Structures & Algorithms, Discrete Mathematics, Calculus III, Intro to Statistics and Data Analysis

EXPERIENCE

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.

Vylet

May 2026 – Present | vyletdata.com | GitHub

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three paying clients

- Launched Vylet, automating a 30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Diagnosed a name-collision defect in ownership-verification logic that was incorrectly rejecting valid acquisition targets sharing a name with an unrelated business elsewhere; the fix lifted the pipeline's lead-qualification rate from 79% to 89% with zero change to sourcing volume.
- Architected a custom asyncpg Data Access Layer storing Gemini embeddings alongside source records; engineered injection-safe SQL timestamp validation that detects stale entries and triggers automatic re-scrapes, keeping the lead database fresh without manual intervention.

PROJECTS

Granular Synthesizer Plugin | C++, JUCE, VST3, AU

github.com/Verdent06/granular-synth

Real-time audio DSP plugin built from scratch in C++/JUCE — sine oscillator through full granular engine with lock-free threading and release-ready VST3/AU binaries

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic `shared_ptr` swap so the UI thread never blocks `processBlock()`.

SignalWeaver | Python, FastAPI, pgvector, Docker

github.com/Verdent06/SignalWeaver

- Implemented semantic search over stored financial news with pgvector cosine similarity (768-d MPNet embeddings), hitting 49ms p50 / 99ms p99 over 90 queries returning top-5 or top-10 matches.
- Lifted financial-sentiment classification accuracy from 81% to 96% by LoRA fine-tuning a quantized meta-llama/Meta-Llama-3.1-8B-Instruct model on 3,454 Financial PhraseBank entries, evaluated on a held-out test set.
- Containerized the stack with Docker Compose (API + Postgres/pgvector + nginx frontend) and a GitHub Actions CI pipeline (frontend build, pytest, API image build on main).

TECHNICAL SKILLS

Languages Python, C++, SQL

Frameworks FastAPI, RxJS, JUCE

Databases pgvector, Redis

Cloud & Infrastructure Docker, Celery, GitHub Actions, Git